SIMATS ENGINEERING

ASSESSMENT TOOL 2: Research Review & Analysis

COURSE NAME: ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS **COURSE CODE: MLA0101**

COURSE OUTCOME COVERED
CO4: *Develop intelligent software agents using suitable agent architectures and learning techniques.* (BTL 3)
Assessment Tool Weightage: 30%

Total Marks: 100 **Time: 90 Mins**
No. of Questions: 1

Annexure A

Research Review & Analysis

Code of Conduct:
I **Kruthika Sre S / 192525360** *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*
Signature of the student: Kruthika Sre

Criterion	Weightage	Marks Obtained
1. Problem Analysis and Domain Understanding	10%	
2. Literature Search and Source Selection	10%	
3. Literature Review and Synthesis	15%	
4. Comparative Analysis of AI Techniques	10%	
5. Critical Analysis of Research Findings	10%	
6. Research Gap Identification	10%	
7. Proposed Research Direction	10%	
8. Research Methodology / Analysis Framework	5%	
9. Experimental / Evidence-Based Validation	10%	
10. Result Analysis and Interpretation	5%	
11. Research Paper Quality and Referencing	10%	
12. Technical Presentation and Research Defense	5%	
Total	100%	

Signature of the Faculty Total Marks Awarded

ANSWER ANY ONE QUESTION

S.No.	Research Review & Analysis Question	Primary Topic
1	**Decision Tree Learning:** Review recent research on ID3, C4.5, and CART. Compare their attribute-selection methods, pruning strategies, interpretability, computational complexity, and classification performance. Identify a research gap and suggest a possible improvement.	Decision Trees
2	**Neural Networks vs Statistical Learning:** Review and compare research on neural-network and statistical-learning approaches for prediction and classification. Analyze their performance, data requirements, generalization, interpretability, computational cost, and suitability for different applications.	Statistical Learning & Neural Networks
3	**Kernel Methods:** Analyze research on SVM and kernel methods for non-linear classification. Compare linear, polynomial, and RBF kernels and evaluate their advantages and limitations compared with neural-network-based approaches.	Kernel Methods
4	**Reinforcement Learning:** Review research on Q-Learning, SARSA, and Deep Q-Networks. Compare their learning mechanisms, exploration strategies, convergence, computational requirements, and real-world applications. Identify challenges and research gaps.	Reinforcement Learning
5	**Explainable and Generalizable AI:** Review research on Inductive Learning and Explanation-Based Learning and analyze how these approaches contribute to generalization and explainability in AI. Identify current limitations and propose future research directions.	Inductive & Explanation-Based Learning

Structure of the Report:

S.No.	Paper Section	Expected Content
1	**Title**	Specific and concise title related to the selected research question/topic.
2	**Abstract**	Brief summary of the research problem, review methodology, major findings, research gap, and conclusion.
3	**Keywords**	4–6 important keywords related to the selected topic.
4	**1. Introduction**	Introduce the AI topic, background, importance, motivation, and scope of the review.
5	**2. Research Problem and Objectives**	Clearly state the research problem and specific objectives of the review.
6	**3. Literature Review**	Review relevant research papers and summarize major approaches, methodologies, findings, and contributions.
7	**4. Comparative Analysis**	Compare the reviewed approaches based on suitable parameters such as accuracy, complexity, interpretability, computational cost, scalability, and generalization.
8	**5. Critical Analysis**	Discuss strengths, weaknesses, limitations, assumptions, and practical challenges identified across the reviewed studies.
9	**6. Research Gap**	Identify unresolved problems, limitations in existing research, and areas requiring further investigation.
10	**7. Proposed Research Direction**	Suggest possible solutions, improvements, or future research directions based on the identified gap.
11	**8. Discussion**	Discuss the overall insights obtained from the review and their implications for AI research and applications.
12	**9. Conclusion**	Summarize the major findings, key insights, research gap, and future direction.
13	**References**	List all research papers, books, datasets, websites, and other sources using a consistent citation style.

Evaluation Rubrics:

Criterion	Weightage (%)	Excellent	Good	Average	Poor
1. Research Problem Identification and Understanding	10%	9–10% – Clearly identifies the research problem, objectives, scope, and relevance to AI learning approaches.	7–8% – Research problem and objectives are clearly identified with minor omissions.	5–6% – Basic understanding of the research problem with some gaps.	0–4% – Problem is unclear, inappropriate, or poorly understood.
2. Literature Search and Source Selection	10%	9–10% – Uses diverse, highly relevant, recent, and credible research papers, journals, conferences, and technical sources.	7–8% – Uses relevant and credible sources with minor limitations.	5–6% – Uses a limited number of relevant sources.	0–4% – Sources are insufficient, unreliable, or largely irrelevant.
3. Literature Review and Synthesis	15%	13–15% – Provides a comprehensive review and synthesizes findings across multiple studies rather than merely summarizing them.	10–12% – Good review with meaningful comparison of major studies.	7–9% – Basic review with limited synthesis and comparison.	0–6% – Superficial review with little understanding of existing research.

4. Analysis of AI Techniques / Methods	15%	13–15% – Provides technically accurate and in-depth analysis of relevant methods such as decision trees, statistical learning, neural networks, kernel methods, or reinforcement learning.	10–12% – Provides good technical analysis with minor gaps.	7–9% – Provides basic technical description with limited analysis.	0–6% – Methods are poorly understood or inaccurately described.
5. Comparative Analysis of Research Approaches	10%	9–10% – Critically compares approaches based on accuracy, complexity, interpretability, scalability, computational cost, and application suitability.	7–8% – Provides meaningful comparison using several parameters.	5–6% – Basic comparison with limited criteria.	0–4% – Little or no meaningful comparison.
6. Critical Evaluation of Research Findings	10%	9–10% – Critically evaluates strengths, weaknesses, assumptions, experimental methods, and findings of existing studies.	7–8% – Provides good critical evaluation with minor gaps.	5–6% – Some critical observations are provided but analysis is limited.	0–4% – Mostly descriptive with little critical evaluation.

7. Identification of Research Gaps	10%	9–10% – Clearly identifies significant research gaps, limitations, unresolved issues, and opportunities for further investigation.	7–8% – Identifies relevant research gaps with reasonable justification.	5–6% – Identifies basic gaps but provides limited justification.	0–4% – Research gaps are missing or incorrectly identified.
8. Research Insights and Recommendations	5%	5% – Provides insightful conclusions and technically feasible recommendations for future research or applications.	4% – Provides relevant conclusions and recommendations.	2–3% – Basic conclusions with limited recommendations.	0–1% – Conclusions or recommendations are absent or inappropriate.
9. Technical Report and Referencing	10%	9–10% – Report is well structured, technically accurate, professionally written, and uses a consistent citation/reference style.	7–8% – Well-structured report with minor writing or referencing issues.	5–6% – Adequate report but contains several structural or referencing issues.	0–4% – Poorly structured, inadequately referenced, or incomplete report.
10. Originality, Academic Integrity and Use of Sources	5%	5% – Demonstrates original synthesis, proper citation, and strong academic integrity with no inappropriate copying.	4% – Good originality and proper citation with minor issues.	2–3% – Some original analysis but significant dependence on sources.	0–1% – Evidence of substantial copying, poor citation, or academic-integrity concerns.

1. DECISION TREE

A Comparative Review of ID3, C4.5, and CART Decision Tree Learning: Attribute Selection, Pruning, Interpretability, and Classification Performance

Abstract

Decision trees are widely used supervised learning techniques because of their simplicity, interpretability, and ability to represent classification decisions as a sequence of rules. This review compares three foundational algorithms: ID3, C4.5, and CART. The comparison focuses on attribute-selection methods, pruning strategies, computational complexity, interpretability, and classification performance. ID3 uses entropy and information gain, C4.5 improves upon ID3 through gain ratio, continuous-attribute handling, and error-based pruning, while CART constructs binary trees primarily using the Gini impurity criterion and cost-complexity pruning. Recent research continues to show that no single algorithm consistently performs best across all datasets because performance depends on data size, feature type, noise, and class distribution. The review identifies a research gap in developing a unified decision-tree approach that dynamically selects splitting and pruning strategies according to dataset characteristics while maintaining interpretability. A possible research direction is an adaptive hybrid decision tree combining multiple attribute-selection criteria with data-dependent pruning.

Introduction

Decision tree learning is an important machine learning technique used for classification and regression. A decision tree represents decisions through internal nodes, branches, and leaf nodes, making its predictions relatively easy to understand.

ID3, introduced by Ross Quinlan, selects attributes using information gain and primarily handles categorical attributes. C4.5 was developed as an extension of ID3 and introduced gain ratio, continuous-attribute handling, and pruning. CART constructs binary trees and is commonly associated with the Gini impurity criterion for classification.

Recent research continues to examine decision trees because they provide an attractive balance between predictive performance and interpretability, particularly in applications such as medical diagnosis and fraud detection.

Research Problem and Objectives

Research Problem:

Although ID3, C4.5, and CART are well-established algorithms, their performance varies depending on dataset characteristics. There is therefore a need to understand their differences and identify opportunities for improving their adaptability, accuracy, and generalization while preserving interpretability.

Objectives:

1. To review the principles of ID3, C4.5, and CART.
2. To compare their attribute-selection methods.
3. To analyze their pruning strategies.
4. To compare interpretability and computational complexity.
5. To examine their classification performance across different dataset conditions.
6. To identify limitations and a research gap.
7. To propose a possible improvement to conventional decision-tree learning.

Literature Review

Quinlan's ID3 introduced entropy-based attribute selection using information gain. Its major limitation is that information gain can favor attributes containing many distinct values. ID3 also has limited support for continuous attributes and does not provide the sophisticated pruning mechanisms found in later algorithms.

C4.5 was developed as an extension of ID3. It uses gain ratio to reduce the bias of information gain toward attributes with many possible values. It can also process continuous attributes and employs post-growing error-based pruning to improve generalization.

CART, or Classification and Regression Trees, constructs binary trees. For classification, Gini impurity is commonly used to select splits, while cost-complexity pruning controls tree size and reduces overfitting. CART can also be extended to regression problems, which distinguishes it from the original ID3 and C4.5 classification-focused approaches.

A 2024 IEEE Access survey reviewed decision-tree development from classical algorithms such as ID3, C4.5, and CART to modern ensemble approaches, emphasizing the continuing importance of interpretability and decision-tree-based learning.

Comparative research also indicates that accuracy, tree size, and computational time can change considerably between algorithms and datasets. Some datasets favor C4.5 or CART, while others can favor ID3, demonstrating that algorithm performance is dataset-dependent.

Comparative Analysis

Criterion	ID3	C4.5	CART
Attribute selection	Information Gain	Gain Ratio	Gini Impurity
Tree structure	Multiway	Multiway	Binary
Categorical data	Yes	Yes	Yes
Continuous data	Limited	Yes	Yes
Pruning	Basic/limited	Error-based post-pruning	Cost-complexity pruning
Main objective	Maximize information gain	Reduce information-gain bias	Minimize impurity
Interpretability	High	High	High
Computational complexity	Relatively simple	Higher than ID3	Efficient binary splitting, but complexity depends on implementation
Regression	No	Primarily classification	Yes
Overfitting control	Weaker	Stronger	Strong
Typical tree size	Can become large	Often more compact	Usually binary and potentially deeper
Main limitation	Bias toward multi-valued attributes	More computational processing	Binary structure can create deeper trees

ID3 is conceptually the simplest of the three, while C4.5 provides several improvements over it. CART's binary structure makes its decision process systematic, although a binary tree can sometimes require greater depth than a multiway tree.

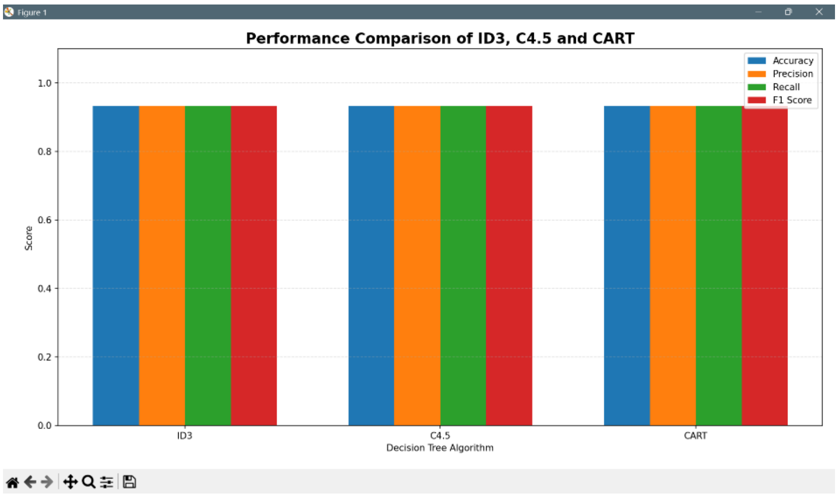
Critical Analysis

ID3 has the advantage of simplicity and relatively straightforward implementation. However, information gain can favor attributes with many values, potentially producing unnecessarily large trees. Its limited treatment of continuous data further restricts its application to modern mixed datasets.

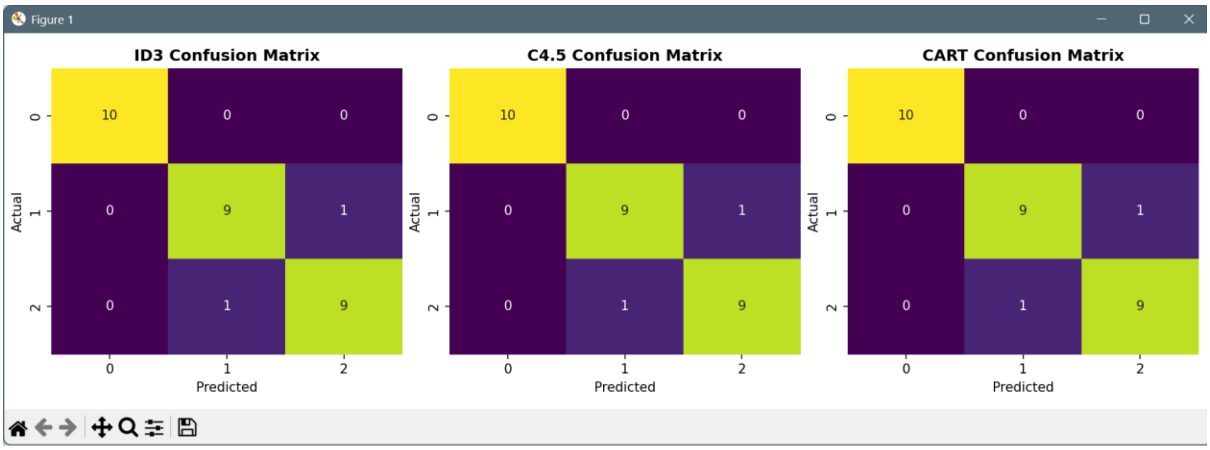
C4.5 addresses several ID3 limitations through gain ratio, continuous-variable handling, and pruning. However, gain ratio itself is not universally optimal because it can favor highly unbalanced splits. Therefore, although C4.5 is more sophisticated, its selection criterion can still introduce bias.

CART provides a strong alternative through binary splitting and Gini-based selection. Its cost-complexity pruning provides a systematic mechanism for controlling tree size. However, the binary-tree restriction may result in deeper structures, potentially affecting interpretability when datasets require many sequential splits.

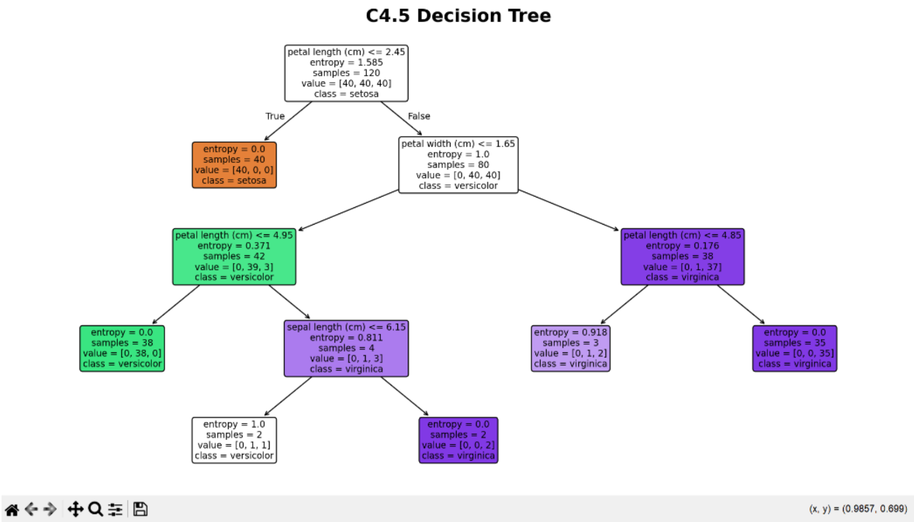
Overall, classification performance cannot be ranked universally as ID3 < C4.5 < CART. Experimental evidence indicates that the best algorithm depends on dataset characteristics, including sample size, attribute types, noise, and class distribution.

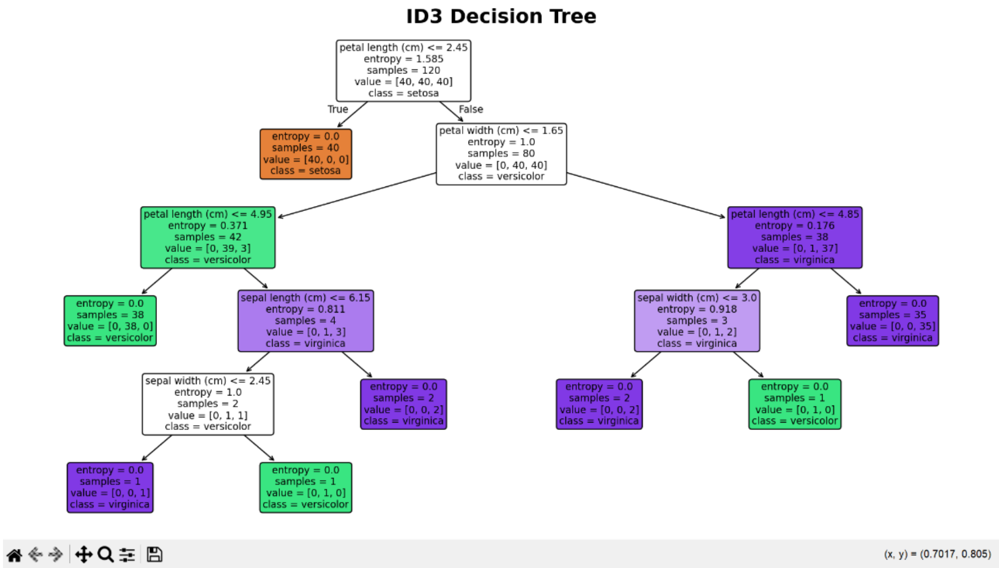


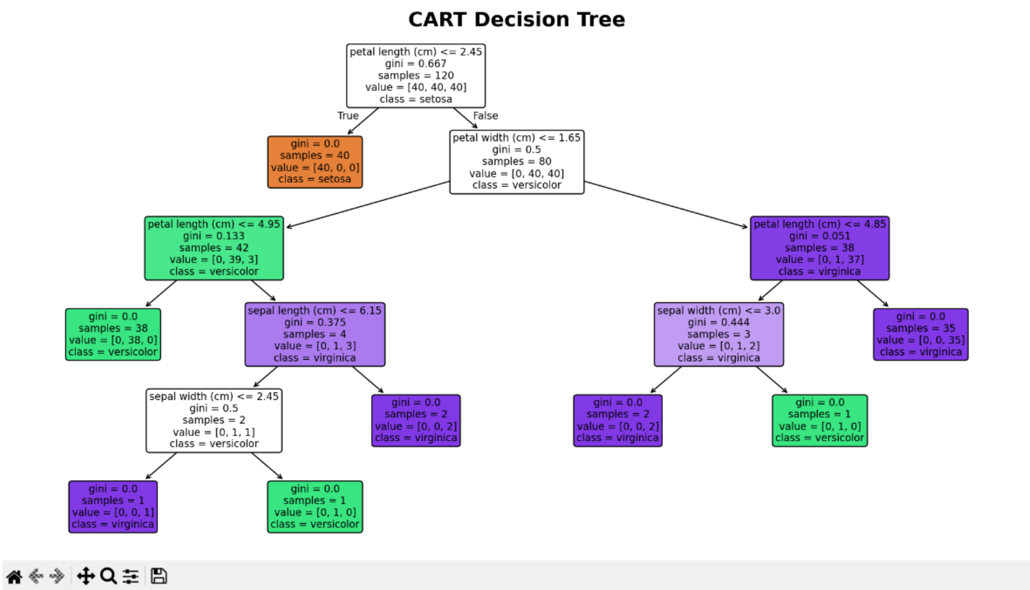
4.1 Performance Comparison



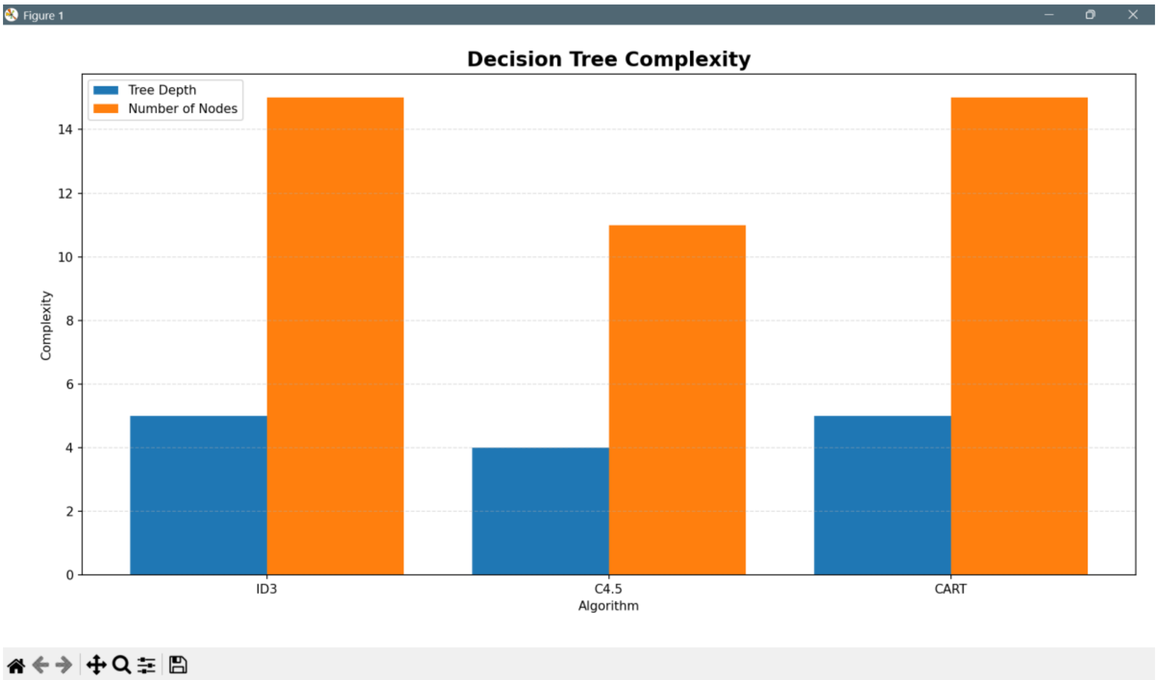
4.2 Confusion Matrix Comparison







4.3 Decision Tree Visualization



4.4 Complexity Comparison

```
======================================================================
       DECISION TREE ALGORITHM COMPARISON
======================================================================
Algorithm  Accuracy  Precision   Recall  F1 Score  Tree Depth  Nodes
      ID3  0.933333   0.933333 0.933333  0.933333           5     15
     C4.5  0.933333   0.933333 0.933333  0.933333           4     11
     CART  0.933333   0.933333 0.933333  0.933333           5     15

======================================================================
BEST PERFORMING ALGORITHM: ID3
BEST ACCURACY: 93.33%
======================================================================
```

4.5 Overall Results

Research Gap

Existing studies provide extensive comparisons of ID3, C4.5, and CART, but many comparisons treat the splitting criterion and pruning strategy as fixed characteristics of each algorithm.

A significant research opportunity is therefore the development of an adaptive decision-tree framework that dynamically chooses the most appropriate attribute-selection criterion and pruning strategy based on the characteristics of the input dataset.

Such a framework could address the limitations of relying on a single criterion such as information gain, gain ratio, or Gini impurity.

Proposed Research Direction

A possible improvement is an Adaptive Hybrid Decision Tree (AHDT).

The proposed approach could:

1. Analyze dataset characteristics such as number of samples, feature types, class imbalance, and number of unique attribute values.
2. Dynamically select between information gain, gain ratio, and Gini impurity.
3. Generate candidate trees using the selected criteria.
4. Apply data-dependent pruning rather than using a fixed pruning strategy.
5. Compare candidate trees using accuracy, F1-score, tree depth, number of nodes, and computational time.
6. Select the smallest interpretable tree that achieves competitive classification performance.

This approach could combine the strengths of ID3, C4.5, and CART while reducing their individual weaknesses.

Discussion

The comparison shows that decision-tree learning involves a trade-off between simplicity, computational efficiency, generalization, and predictive performance. ID3 provides a simple foundation, C4.5 improves attribute selection and pruning, and CART provides flexible binary classification and regression capabilities.

Recent research also demonstrates that decision trees remain relevant despite the growth of ensemble methods. Their interpretability makes them particularly valuable when users need to understand why a prediction was produced.

An adaptive approach could therefore be more practical than declaring one classical algorithm universally superior. Future experiments should evaluate the proposed method on datasets with different sizes, feature types, noise levels, and class distributions.

Conclusion

ID3, C4.5, and CART represent three important stages in the development of decision-tree learning. ID3 uses information gain, C4.5 introduces gain ratio and improved pruning, and CART uses binary splitting with Gini impurity and cost-complexity pruning. Each algorithm provides different advantages in terms of simplicity, computational requirements, interpretability, and classification performance.

The review identifies the lack of a universally optimal splitting and pruning strategy as an important research opportunity. An adaptive hybrid decision tree that dynamically selects these strategies could improve classification performance while retaining the transparency that makes decision trees valuable.

References

1. Quinlan, J. R. (1986). Induction of Decision Trees. Machine Learning, 1, 81–106.
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3. Breiman, L., Friedman, J. H., Olshen, R. A., & Stone, C. J. (1984). Classification and Regression Trees. Wadsworth.
4. Mienye, I. D., & Jere, N. (2024). A Survey of Decision Trees: Concepts, Algorithms, and Applications. IEEE Access, 12, 86716–86727.
5. Huang, Y., et al. (2021). Splitting Choice and Computational Complexity Analysis of Decision Trees. Entropy, 23(10), 1241.
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